JOURNAL

the New eVTOL to change the market



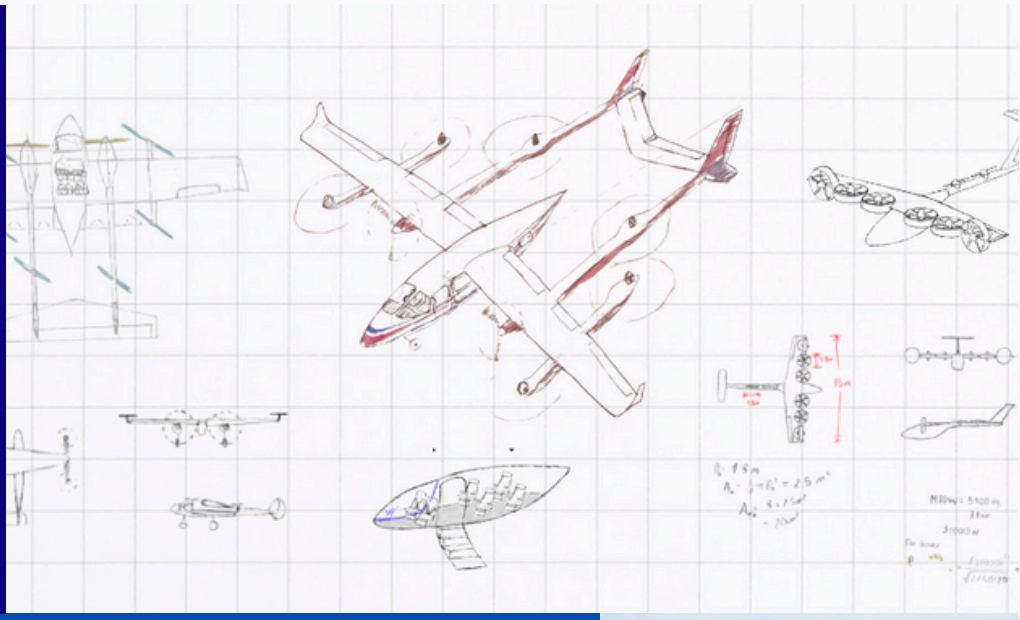
Conceptual Design

Intriguing facts, insights, experiences... The most interesting of this design.

An overview of the design process

Napkin design

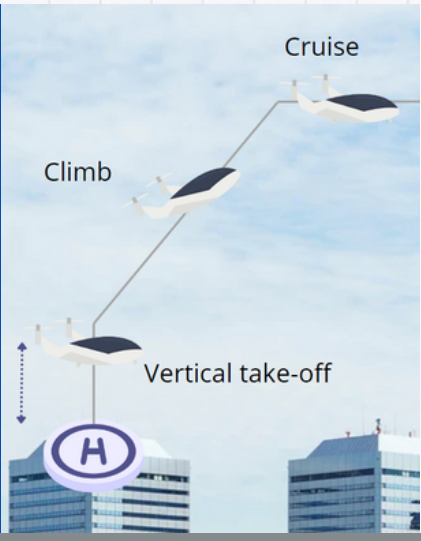
Every project begins with simple sketches and ideas, allowing creativity to take the lead. As the design process progresses, these initial concepts are refined and validated through the application of calculus and physics, ensuring their feasibility and functionality.





Intended for in-city operations that allow for multiple flights during the day. Entering into the **enhanced** category according to EASA.

Especially inspired for the European market, where there are many major cities that can be easily connected with these aircraft. It is able to adapt to an urban environment with many buildings and few airfields with runways suitable for conventional take-off and landing.



The eVTOL concept has gained popularity in the last few years, and several companies have already developed their own prototypes, and have even carried out the first test flights.

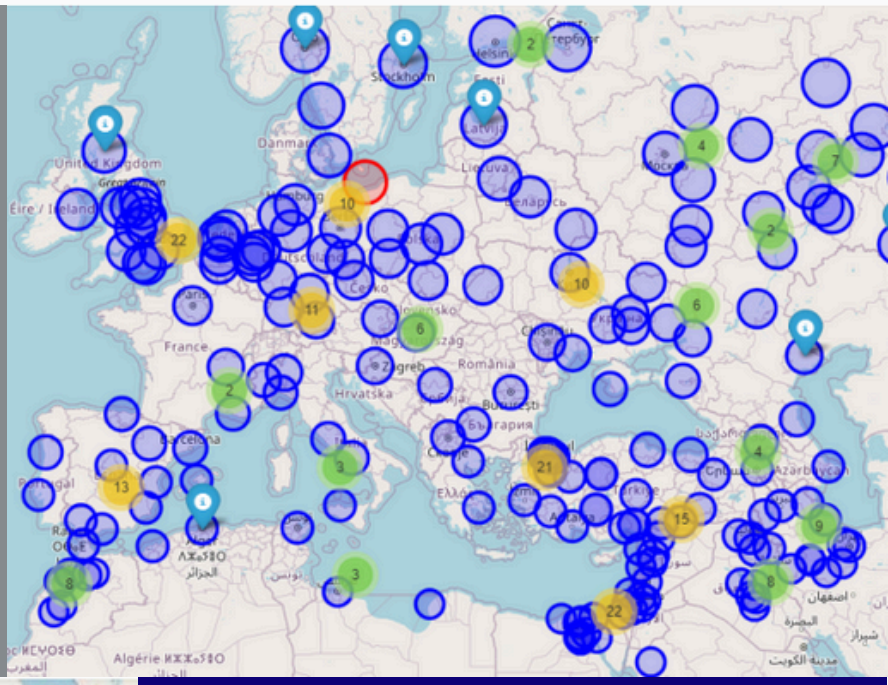
But to get to that point they have taken years and a lot of investment behind these ambitious projects.



These vehicles are not only intended to improve urban mobility but also to provide zero carbon solutions as most of them are designed with electric motors.

Market opportunity

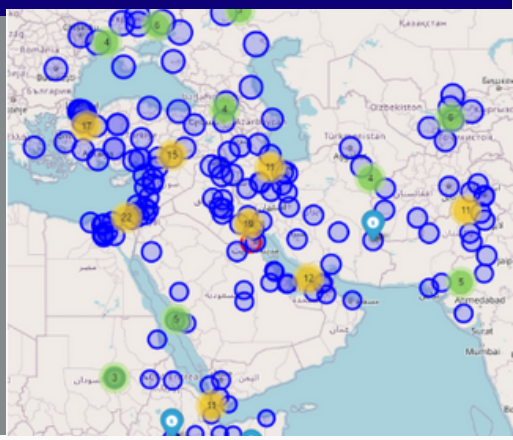
Europe's landscape reveals a dense network of potential eVTOL markets, with the highest concentrations appearing in the Western and Central regions. Countries like the United Kingdom, France, Germany, Italy, and Spain demonstrate significant coverage, with urban areas well within the 100 km operational range. This indicates that many European cities could benefit from eVTOL connectivity, enhancing intra-country and cross-border mobility.





In the United States, there is a noticeable concentration of potential markets along the East Coast, spanning from the Boston-Washington D.C. corridor to Florida, indicating a high demand for eVTOL services in densely populated areas. The West Coast shows potential markets focused around urban centers like Los Angeles, San Francisco, and Seattle. Inland cities such as Denver and Phoenix also emerge as key areas where eVTOLs could significantly impact transportation dynamics.

The Middle East presents an emerging market with urban centers such as Dubai, Riyadh, and Doha poised to adopt eVTOL technologies. The distribution suggests that the range capabilities align well with the geographic spacing of cities in this region, potentially transforming travel and logistics in an area that is rapidly urbanizing and investing in technological advancements.

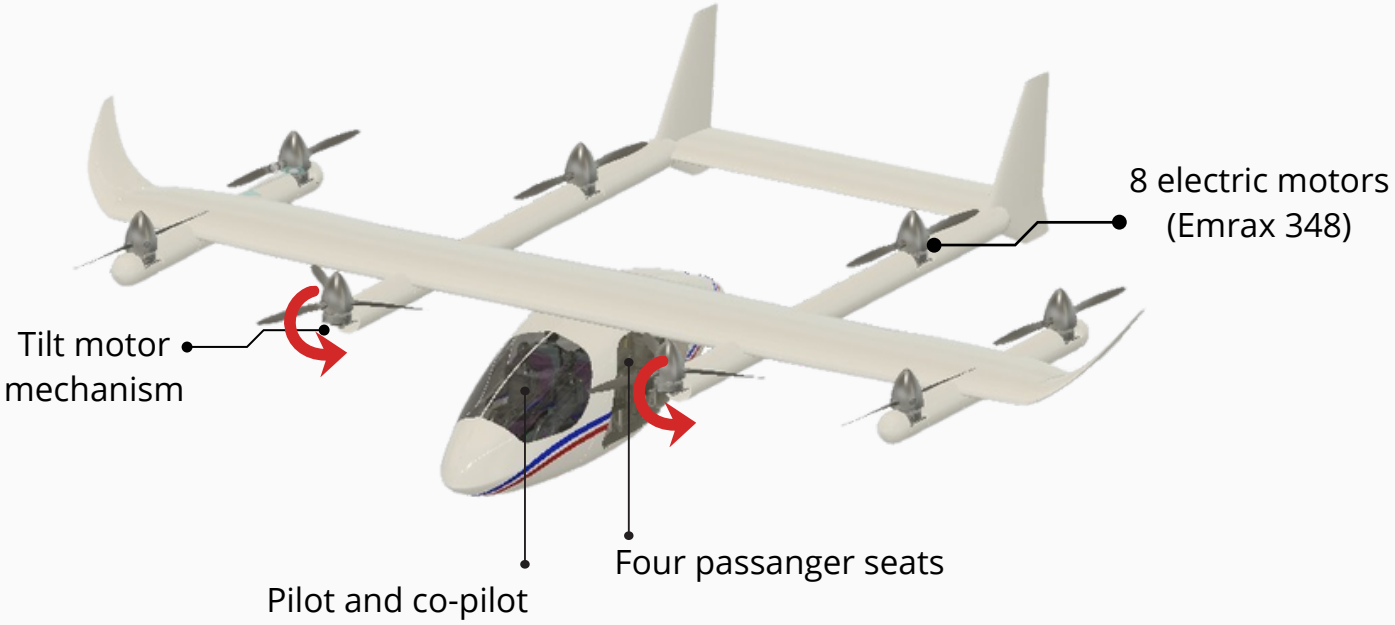


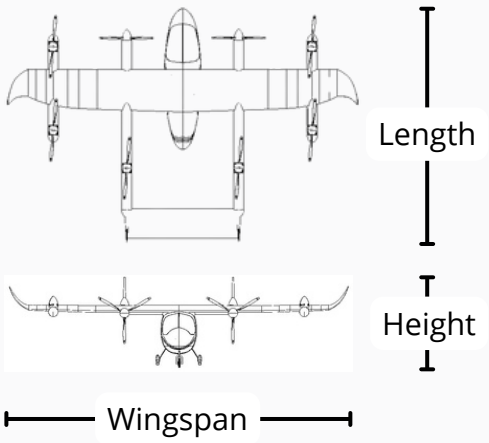
Technical data

The design represents a cutting-edge concept, featuring a combination of six fixed-position engines for takeoff, landing, and low-speed maneuvering. These six engines, equipped with two-blade propellers, are aligned along the longitudinal axis to minimize drag during flight stages where lift is provided by the wings.

Additionally, the aircraft is equipped with two tilting electric engines utilizing three-blade propellers. This configuration optimizes the thrust-to-weight ratio during vertical takeoff and landing, ensuring enhanced safety and redundancy with a total of eight engines. During horizontal flight, these tilting engines transition to provide thrust while the wings generate lift.

The pilot and copilot positions are designed to offer the best possible visibility. The tilt-rotor mechanism is strategically placed to optimize lift distribution and maintain a clear area for passenger circulation, fully complying with aviation regulations.





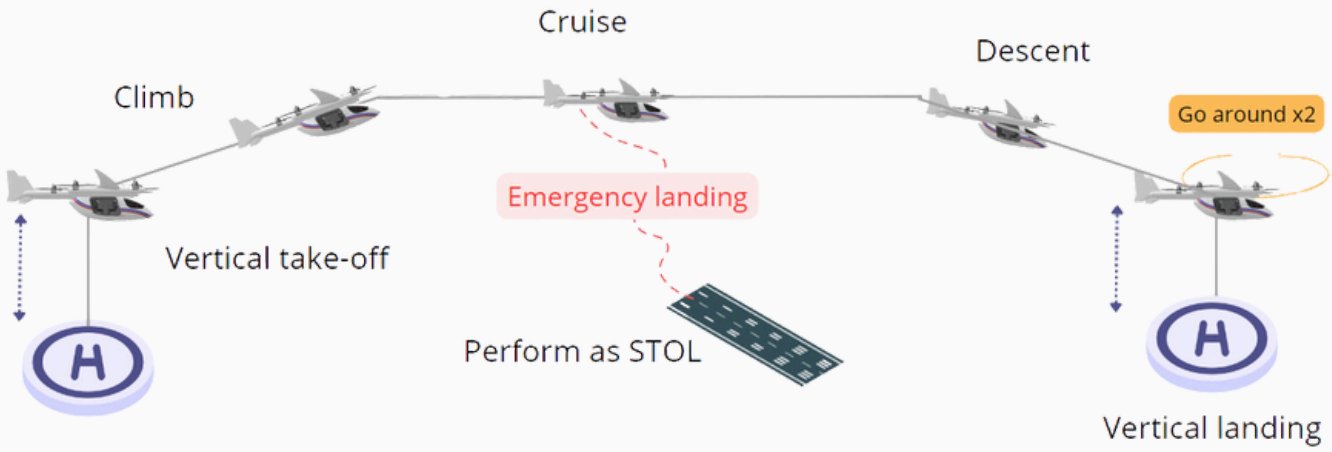


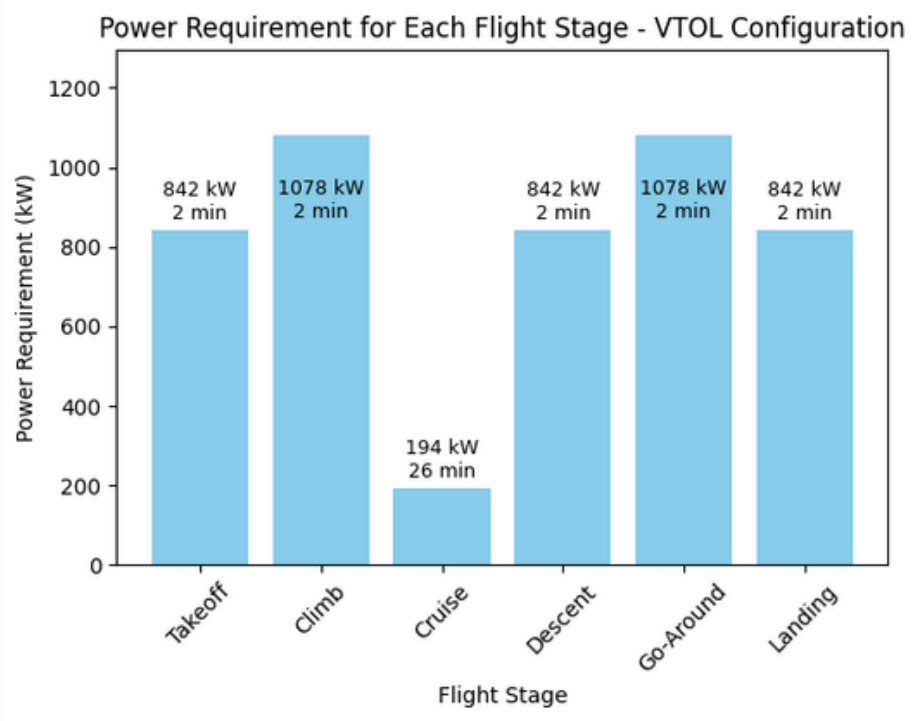
Cruise speed:	248 km/h - 134 knots
Range*:	100 km - 62 mi
Payload:**	600 kg - 1323 lb
Wing span:	13.2 m - 43'4"
Length	9.55 m - 31'4"
Height:	3.6 m - 11'10"

*Including reserve **6 passangers and their baggage

Mission profile

The primary focus is on passenger service, which is why the enhanced operation for dense urban environments was a guiding principle in the design. This allows the aircraft to efficiently serve modern cities, connecting business districts to airports and other tourist attractions, while ensuring safety during emergency landings on vertiports located atop buildings. Additionally, the aircraft is capable of executing emergency landings at short STOL airports.





A preliminary estimate is made of how long each stage can take and the estimated consumption time. It is observed that the stage with the highest consumption will be the takeoff and vertical landing, although they last the shortest time, these stages require a peak discharge at a high rate.

The initial estimation of energy and power only takes into account the desired performance, the detailed transition mechanism analysis will provide more information.

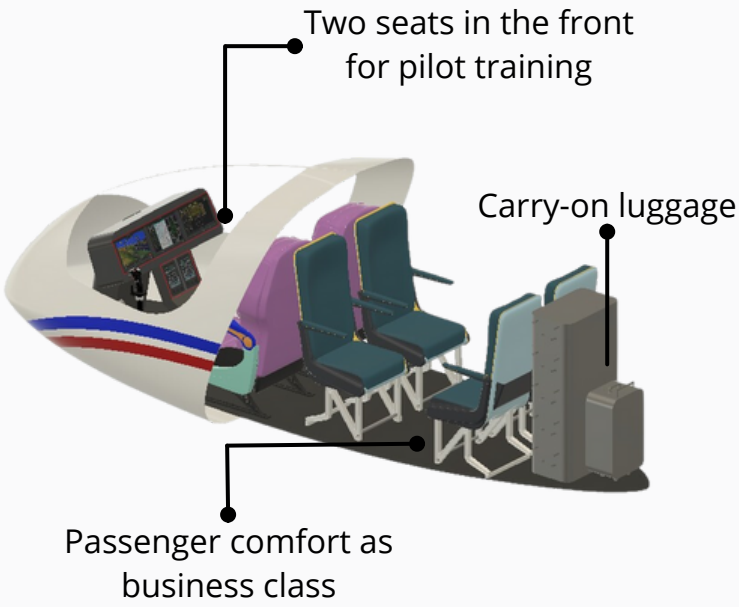
Technical data



The vertical motors are equipped with two-blade propellers and are locked in place during the cruise phase after ascent, reducing drag.



Tilt engines are those used for cruise flight and have three blades to optimize them for this condition.



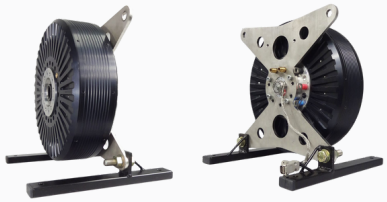
The passenger cabin concept provides comfort for access, also wide windows to enjoy the view from the air. The spacing where planned to match a business class it is ample and also a volume area is provided for hand luggage at the tail cone.

One of the challenges of this innovative transportation technology is the training of the pilots, besides using top of the line avionics to reduce the workload during the flight (Garmin 3000), we included a place for a co-pilot to get real life experience. this is a must for our safety considerations based on skills and qualifications.



Propulsion system

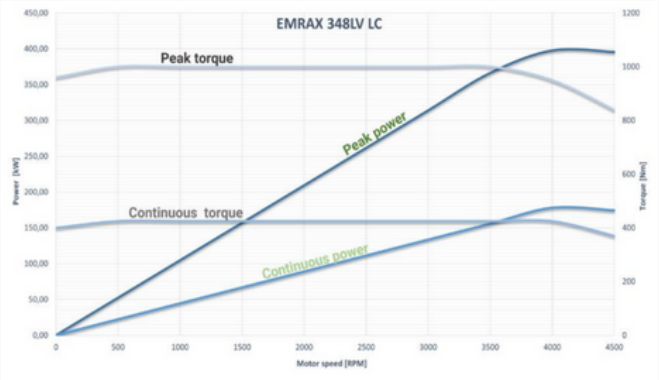


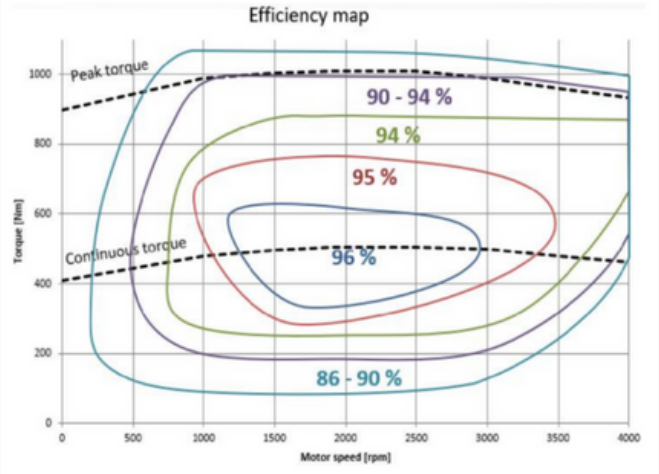


The propulsion system uses high quality motors with peak torque that exceeds many other options on the market. They use liquid cooling and a high thrust to weight ratio, so the safety and margin of maneuver exceed those required by the Civil Aviation Authority.

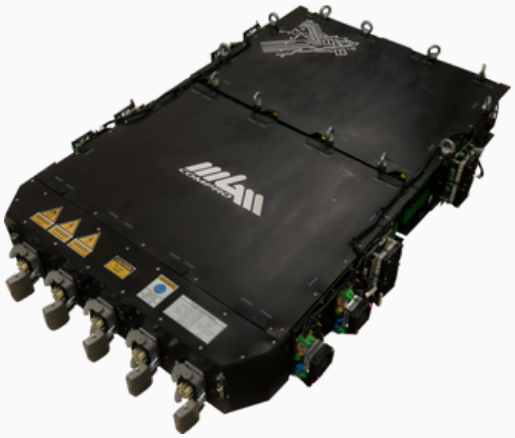
Type:	Axial flux motor
Peak power:	400 kW
Continuous power:	up to 200 kW
Peak torque:	1000 Nm
Continuous torque:	up to 450 Nm
Efficiency:	92-98%

It is a motor produced by Emrax. It was selected for this analysis due to the high availability of information, but other promising companies could provide the aircraft with the solution, such as MGM-COMPRO and magniX.
The Emrax 348 complies with the initial requirements but will not fulfill the requirement of high power and torque at low rpms as shown in graph. The benefit of using an axial flow motor is its **high efficiency**.





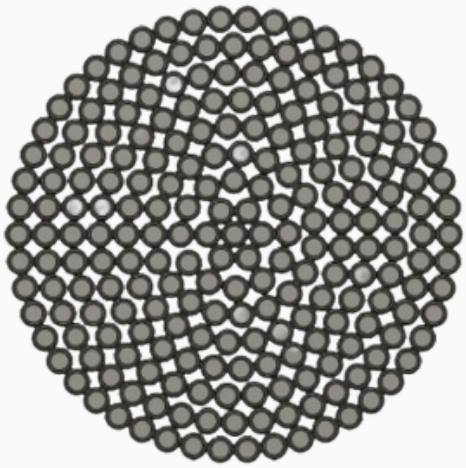
Energy system

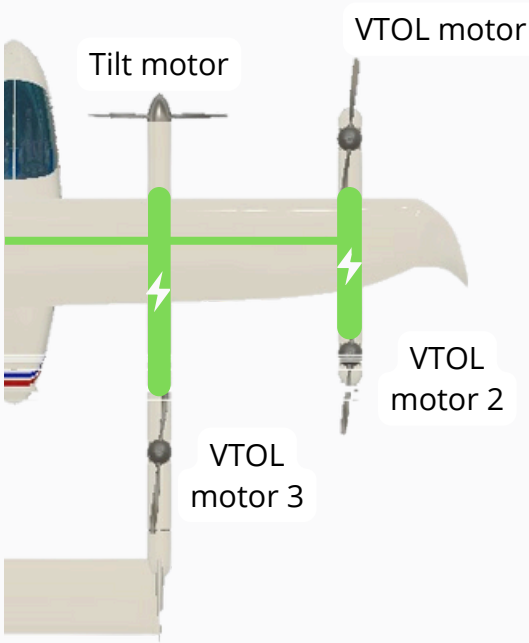


Courtesy: MGM COMPRO

The Li-Ion battery system is currently the best available energy storage technology, renowned for its high energy density and ability to deliver high currents. The battery packs are composed of individual cells that can be arranged in almost any shape to suit the system's requirements. We partnered the MGM COMPRO team that is able to provide the optimal solution with the balance between energy density and C-rate.

Cylindrical battery packs that fit inside the booms were considered. This would allow not only a help in weight and balance of the aircraft but also a saving in long cables to connect the batteries to the engines.

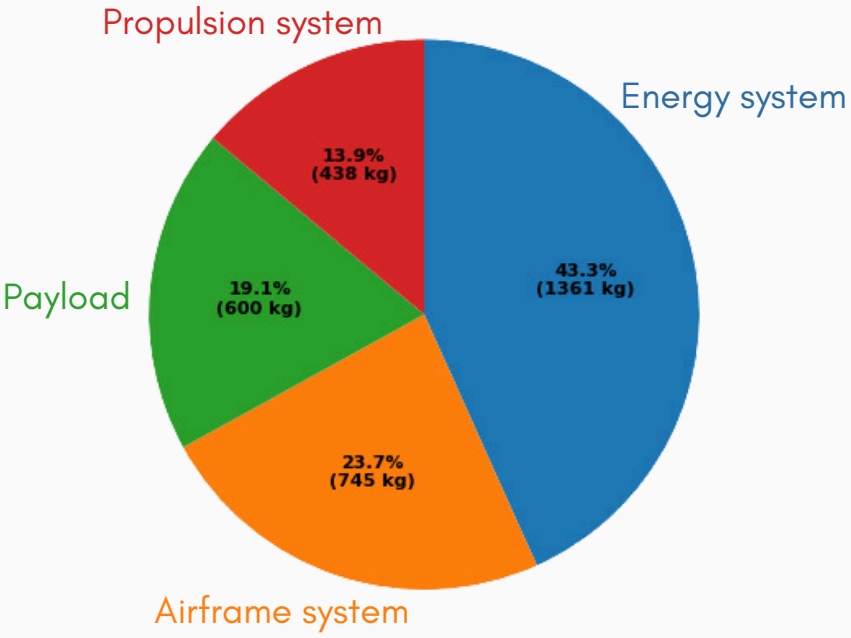


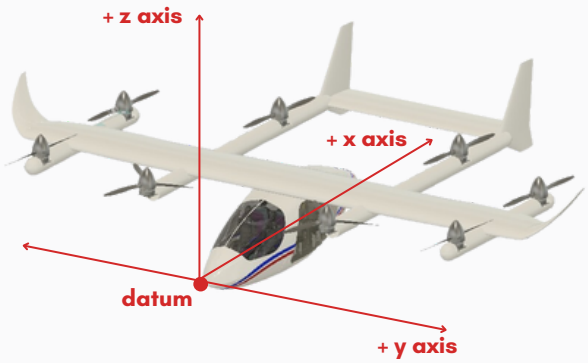


The complete battery pack is divided into smaller packs that will go in each of the booms, corresponding to 2 packs per boom. The idea is that each engine will have its own pack and in case of failure it will be necessary that a single pack can feed 2 engines.

Weight distribution

One of the most crucial aspects of aircraft design is weight distribution, as it significantly affects the stability and control of the aircraft. Initially, only rough estimates can be made due to the lack of detailed designs for each system. However, as the design progresses, iterative refinements allow the results to become increasingly realistic and accurate.





Total weight:

3144 kg

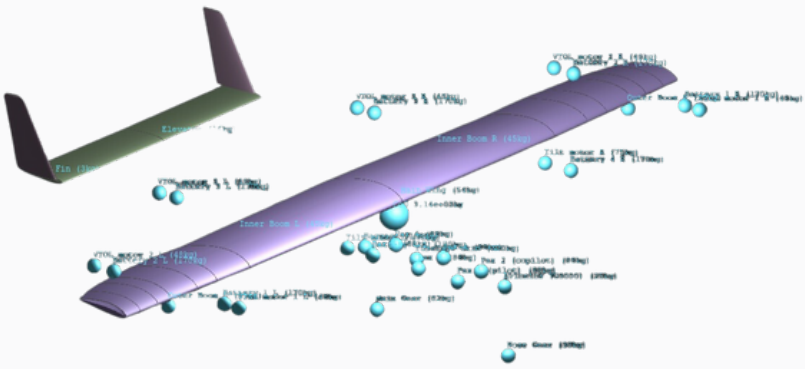
EASA weight limit

3175 kg

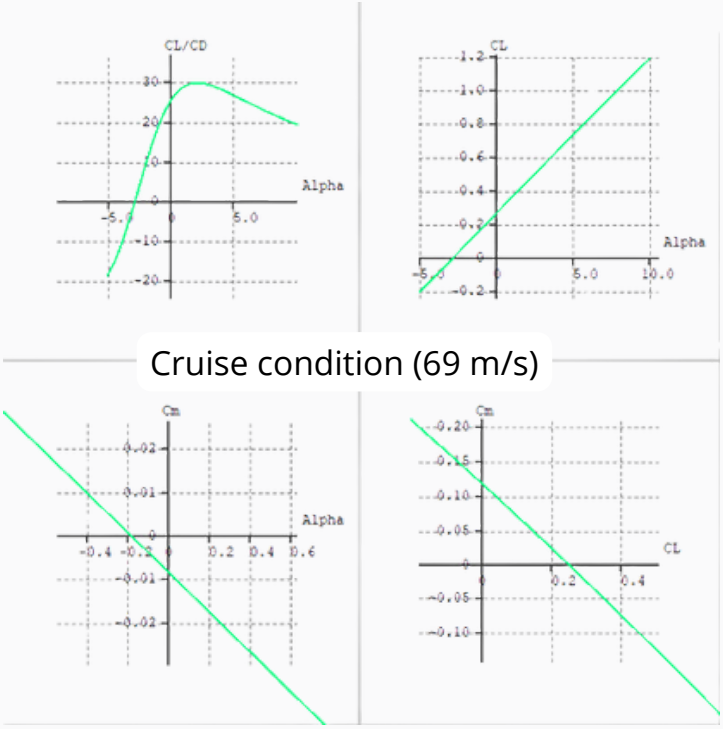
with respect to the datum

Element	Quantity	Weight per unit (kg)	Weight (kg)	x axis in mm	y axis in mm	z axis in mm
Wing	1	54	54	3260	0	1400
Tilt motor	2	75	150	1206	2272	1556
VTOL motor 1	2	48	96	1657	5290	1618
VTOL motor 2	2	48	96	5057	5278	1619
VTOL motor 3	2	48	96	6607	2269	1571
Fuselage skin	1	320	320	3038	0	432
Inner Boom	2	45	90	4751	2269	1235
Outer Boom	2	29	58	3357	5277	1276
Pax. pos 1 (pilot + seat)	2	88	176	1800	270	400
Pax. pos 3 (pax + seat)	2	88	176	2698	287	435
Pax. pos 5 (pax + seat)	2	88	176	3790	266	435
Baggage	6	23	138	4242	0	390
H. tail	1	16	16	8864	0	1251
V. tail	2	1.5	3	9031	2269	1658
Nose landing gear	1	30	30	1036	0	-822
Main landing gear	2	41	82	4091	0	-675
Battery pack 1	2	170.125	340.25	2000	5290	1618
Battery pack 2	2	170.125	340.25	4600	5278	1619
Battery pack 3	2	170.125	340.25	6200	2269	1571
Battery pack 4	2	170.125	340.25	1600	2272	1556
Avionics	1	26	26	975	0	400

Performance, stability and control



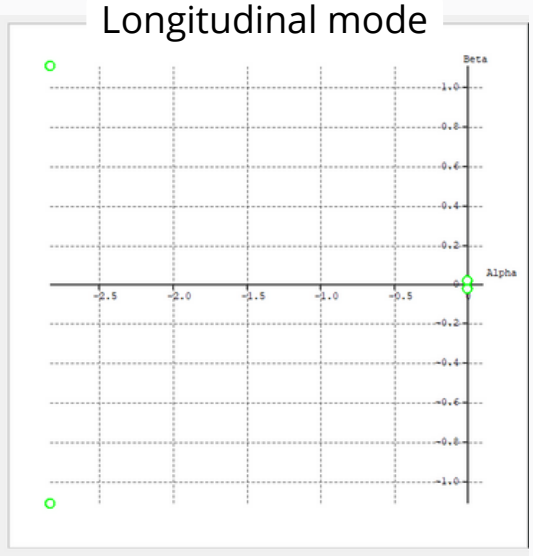
In order to perform the simulation and obtain the stability and control derivatives, a model with the mass distributions must be made, this is done with the XFLR5 tool.



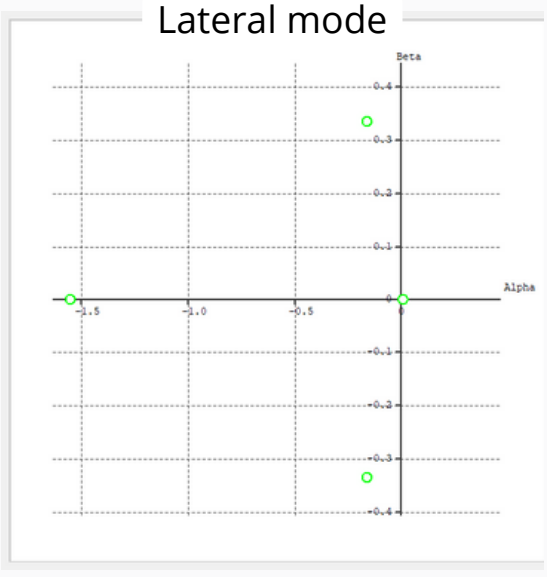
The estimation of the stability of the aircraft was performed using the Ring Vortex Method (VLM2) implemented in XFLR5.

With the mass distribution displayed above, the center of mass sits at a 50% static margin and the behavior of the Cm is acceptable, with Cm vs alpha with negative slope, a CL=0,25 and CL/CD=25 for Cm=0.

Longitudinal mode



Lateral mode



Fact sheet

Parameter	Unit	Value
Performance		
Vmax	m/s	80
VCruise	m/s	69.7
VStall	m/s	31.0
Cruise altitude	ft	1000.0
Aerodynamics		
Wing area	m^2	21.78
Aspect Ratio	-	8.00
Wingspan	m	13.20
Airfoil	-	NACA 4417

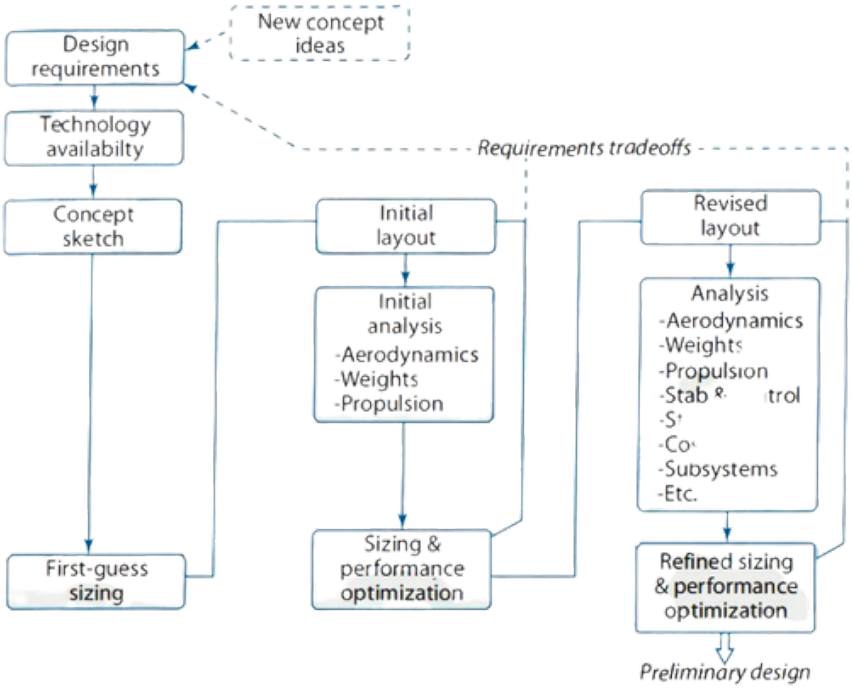
Parameter	Unit	Value
Power and Propulsion		
Motor reference	-	Emrax 348
Motor continuous power	kW	200
Available power VTOL	kW	1600
Available power cruise	kW	400
Motors and Blades		
# Fixed Motors	-	6
Blades (and diameter) fixed motors	-	2 (2.11 m)
# Tilting Motors	-	2
Blades (and diameter) tilting motors	-	3 (1.96 m)

MTOW
3144 Kg

Payload
600 Kg

Battery mass
1361 Kg

Design methodology



The process for achieving this design is adapted from the approaches cited below, using an iterative methodology, where every loop has the result of a better idea of the final design than in the earlier iteration. For this purpose, internal software tools were developed and Fusion360, XFLR5, XFOIL and openVSP were used.
In total, there were 4 iterations of the design to reach this stage of development

- Raymer, D. (2012). *Aircraft design: a conceptual approach*. American Institute of Aeronautics and Astronautics, Inc.
- Gudmundsson, S. (2013). *General aviation aircraft design: Applied Methods and Procedures*. Butterworth-Heinemann.
- Roskam, J. (1985). *Airplane design*. DARcorporation.
- Ugwueze, O., Statheros, T., Horri, N., Bromfield, M. A., & Simo, J. (2023). *An Efficient and Robust Sizing Method for eVTOL Aircraft Configurations in Conceptual Design*. Aerospace, 10(3), 311.

Conclusions and recommendations

- The eVTOL market is determined for significant growth in the coming years, with increasing interest and numerous emerging designs yet to be validated.

- Our design targets strategic markets within a 100 km range, offering the advantage of a substantial payload capacity.

- The aircraft will leverage advancements in battery technology, leading to major performance improvements. Hybrid technology may serve as an intermediate step, bridging the gap until future battery advancements are realized, providing a viable solution in the interim. Evaluating the transition phase effectively and efficiently is crucial for the project's continuation in the next design phases.

- Future stages will require expertise from various fields, along with wind tunnel testing, detailed simulations, and mockups to ensure proper development. The design of a comprehensive control interface, skids for the landing and detailed structural analysis are required.

